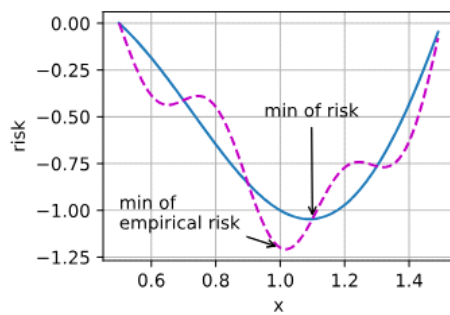
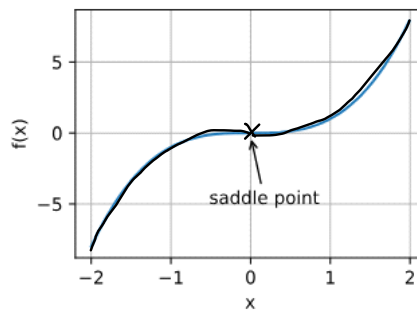
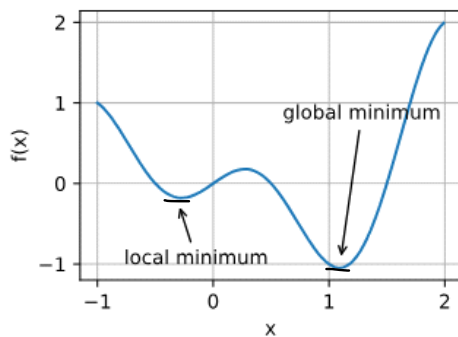


↪ minimization (loss function)

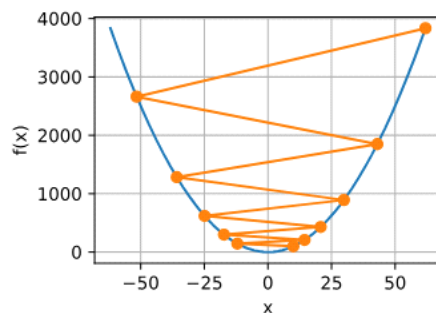
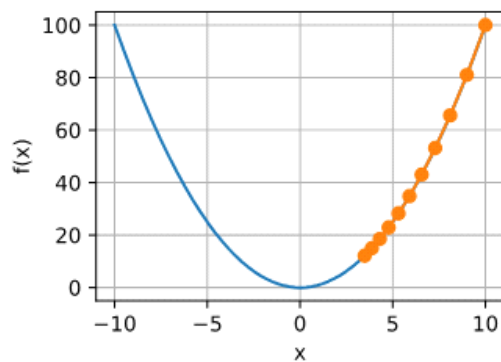


$$f(x) = x \cdot \cos(\pi x) \quad -1 < x < 2$$



$$\hookrightarrow f(x+\epsilon) = f(x) + \epsilon f'(x) + O(\epsilon)^2$$

η

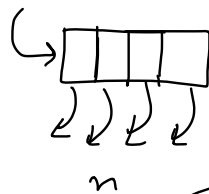


$$x = [x_1, x_2, x_3, \dots, x_d]^T$$

$$f: \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\nabla f(x) = \left[\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_d} \right]^T$$

$$f(x + \epsilon) = f(x) + \epsilon^T \nabla f(x) + \mathcal{O}(\|\epsilon\|^2)$$



$$f_i(x) = \frac{1}{n} \sum_{i=1}^n f_i(x)$$

$$\nabla f(x) = \frac{1}{n} \sum_{i=1}^n \nabla f_i(x)$$

Speed
→ convergence
memory

Gradient vs SGD

↓
slow / noisy

↓
small / faster

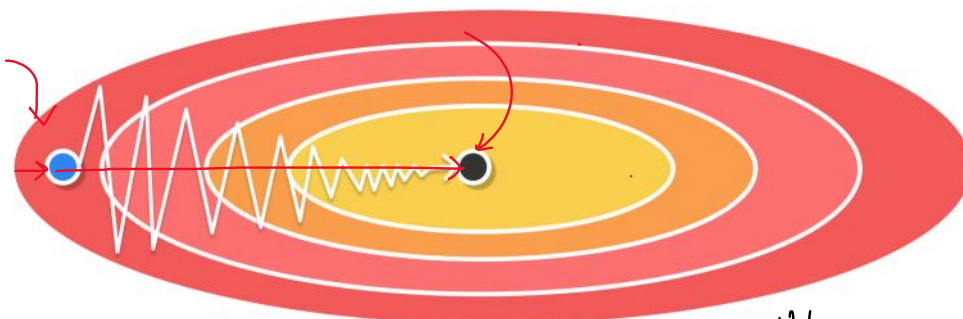
1. higher
- 2.
3. noise

SGD

momentum

adaptive learning rate

$$\underline{w}_t = \underline{w}_{t-1} - \alpha \underline{\delta w}_t$$



$$\underline{w}_t = \beta \underline{w}_{t-1} + (1-\beta) \nabla L(\underline{w}_t)$$

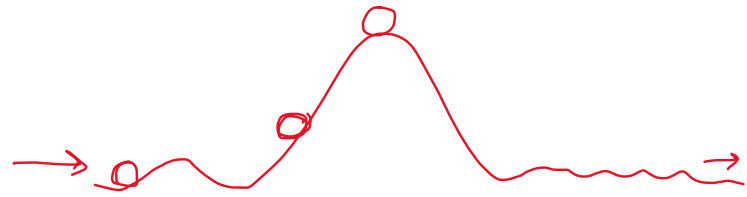
0.5 to 0.9

$w_{1,6}$

(P)

$$\underline{\underline{\theta}}_t = \beta \underline{\underline{\theta}}_{t-1} + (1-\beta) \nabla L(\theta) \quad (\beta)$$

$\theta = \theta - \theta_t$

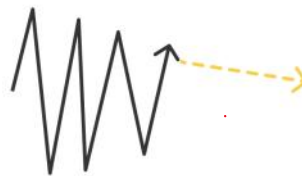


Gradient descent



the gradient computed on the current iteration does not prevent gradient descent from oscillating in the vertical direction

Momentum



the average vector of the horizontal component is aligned towards the minimum

the average vector of the vertical component is close to 0

↳ AdaGrad (η)

exploring $+ \nabla E(w, b) \rightarrow 00$

oscillating $\times$

(w, b) $\nabla E =$

$$\begin{cases} \theta_t = \theta_{t-1} + \eta^2 \\ w_t = w_{t-1} - \frac{\alpha}{\sqrt{\theta_t} + \epsilon} dw_t \end{cases}$$

- ① complexity
- ② parameters

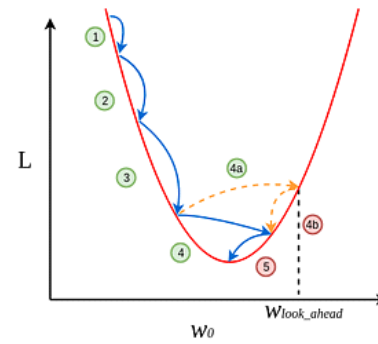
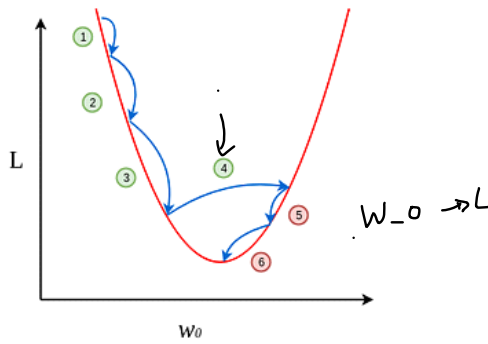
↳ Rms prop (+Adagrad)

$$\begin{aligned} \theta_t &= \beta \theta_{t-1} + (1-\beta) dw_t^2 \\ w_t &= w_{t-1} - \frac{\alpha}{\sqrt{\theta_t} + \epsilon} dw_t \end{aligned}$$

GA
SGD
momentum NA

Nesterov

(2012 - 2022)



(a) Momentum-Based Gradient Descent

(b) Nesterov Accelerated Gradient Descent

$\Rightarrow \frac{\partial L}{\partial w_0} = \frac{\text{Negative}(-)}{\text{Positive}(+)}$

$\Rightarrow \frac{\partial L}{\partial w_0} = \frac{\text{Negative}(-)}{\text{Negative}(-)}$

$\square \rightarrow \text{obj}$

	loss	time	how the model performs
1			
2			
3			
4			
5			
6			

Batch normalization

- 1. faster convergence
 - 2. Reduce sensitivity to initialization
 - 3. Generalization.
- Random

Dropout

- 1. Reduce overfitting
- 2. Robust

- [1. Kettle Creek
2. Round